

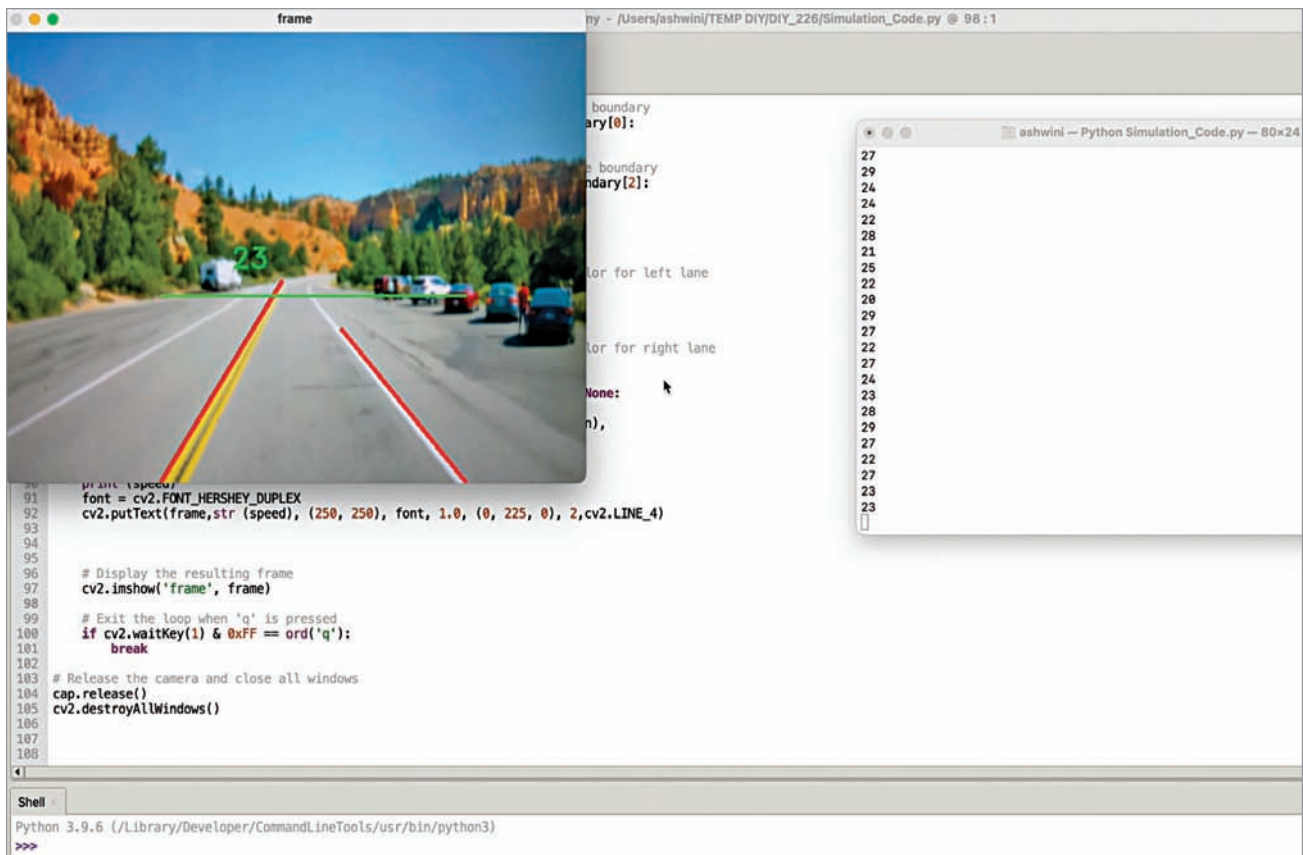


Fig. 10: Code detecting path, lane displaying the speed of approaching vehicles, and the safe distance line marked in green

If these objects are stationary or moving at a constant velocity relative to the vehicle, there is no significant change in the frequency of the returning radar signal.

Doppler shift measurement.

If an object is moving towards or away from the vehicle, it induces a doppler shift in the reflected radar signal. If an object is approaching, the radar signal experiences a positive doppler shift (an increase in frequency). If an object is moving away, it results in a negative doppler shift (a decrease in frequency).

So, here in the doppler radar, we use this principle to measure the speed, which varies in the ADAS as per three different scenarios (see Fig. 3).

Case 1. If the vehicle with radar is stationary and not moving,

and any pedestrian or car comes towards or goes away from that car, then the speed it shows is the speed of the car or of the pedestrian coming or going away from the car with radar.

Case 2. If the car with radar is moving, and there is a non-moving car, people, or a wall in front of it, then it shows the speed at which the car is moving towards/away from that stationary car/person/wall.

Case 3. If the car is moving towards another car, then it shows the speed at which they are going to collide with each other.

If both cars are moving in the same direction, then it shows the difference in their speeds. If both are moving in the same direction with the same speed, then the resulting speed it shows is 0, mean-

ing both are not going to collide. If one of them is moving away faster, then it shows the speed at which the car is going away from the other car. Fig. 4 shows all three scenarios.

If the car in front of it presses the brake or slows down while the blue car behind it keeps moving or increases the speed, then the ADAS display (Fig. 5) shows the resultant speed at which the car in the rear might collide, and the driver needs to maintain the distance marked in the green line in the display of the ADAS to avoid an accident. Alternately, if the driver matches the speed of the car in the front, the relative speed is displayed as 0, meaning both are moving at the same speed with respect to each other, and they will not collide.